BCRP Corpus: An AI-Assisted Research and Information Retrieval System for Central Banking

Gonzalo Bueno Bustíos

12 November 2026

Abstract

We study the use of page-grounded retrieval to support research across public publications of the Banco Central de Reserva del Perú (BCRP). The system links retrieved passages to physical pages in specific PDF artifacts and lets a conversational client restrict its search to a report edition. In a fixed-query comparison on an October 2026 snapshot, hybrid retrieval with page-context reranking returned directly useful first passages for three of six natural-language questions. Keyword, semantic and hybrid retrieval each returned zero under the same wording. Median local retrieval time was 31.609 seconds with reranking and 2.393 seconds for hybrid search, measured over eight and twenty calls, respectively. A separate source audit identified two critical failures among sixteen conversational answers, including a factor-of-ten transcription error. In a later evaluation of instructions that explicitly required source and scope checks, eight of eight reserved cases passed at the answer level. The later exercise had no matched held-out control and lacked raw tool traces. The results illustrate where page-level provenance helps researchers inspect evidence and where judgments about meaning, scope and uncertainty remain necessary.

1. Introduction

A central-bank research question can require evidence of several kinds. A working paper may estimate the response of a variable to an identified shock, a policy report may project that variable under a baseline scenario, and a box in the same report may explain a mechanism using a simulation. These sources can share vocabulary while addressing different populations, horizons and counterfactuals. Retrieval must help a researcher locate the relevant passage and establish which inference it can support.

This paper asks how far page-grounded retrieval can support that task in a local corpus of BCRP publications. We study two observable outcomes: whether a retrieval configuration ranks substantive evidence first, and whether an answer preserves the scope and meaning of the evidence it cites. A passage-ranking experiment compares retrieval configurations under fixed query wording. Two answer studies examine the additional judgments made when a conversational client combines retrieved material into a research answer. The three exercises use different samples and answer different questions.

BCRP Corpus brings two requirements into a single research workflow: trace a claim to a page in a specific PDF, and restrict the search to the edition or information set relevant to the question. The distinction matters when reconstructing a historical policy assessment, since a later report may revise an estimate or the currently stored PDF may postdate the cutoff. The empirical contribution is to document where this workflow found usable evidence and where errors survived even when the relevant page was located. The project is an independent prototype for personal and internal research and carries no BCRP endorsement.[1]

2. Related literature

Korinek (2023) surveys uses of large language models in economics, including background research, data analysis and coding. Central-bank applications address more specific retrieval tasks. Petracca et al. (2026) use retrieval-augmented generation (RAG) to help users find series in the ECB Data Portal, and Seah et al. (2026) build a prototype for exploring economic datasets and their metadata. Gambacorta et al. (2024) train domain-specific language models on central-bank texts and evaluate tasks including monetary policy stance classification. BCRP Corpus addresses a complementary task: finding and inspecting documentary evidence for an economic claim, with explicit edition and physical-page scope.

In finance, *FinanceBench* evaluates open-book answers against evidence from company filings (Islam et al., 2023). Experiments with long-context models show that performance can remain sensitive to the location of relevant material in the input (Liu et al., 2024). These findings motivate measuring retrieval and answer quality separately and testing full-document prompting as a baseline in future work. This paper examines published policy and research documents, where a figure can change meaning across editions, samples or forecast horizons.

[1] The public BCRP Corpus repository contains the portable keyword-search runtime.

3. Data and methods

3.1. Corpus and unit of evidence

All source material indexed and used as documentary evidence in this study came from publicly available BCRP publications. The 11 November 2026 snapshot contains 474 Working Papers from the discovered 1997–2026 inventory, 788 indexed articles from issues 131–207 of *Revista Moneda*, and 89 Spanish *Reportes de Inflación* from June 2002 to September 2026. One *Moneda* record is excluded. Of 555 catalogue entries for inflation report boxes (*recuadros*), 90 have independently searchable bodies. The other 465 retain searchable metadata, with body retrieval through the parent report. There are 1,441 indexed document records, 30,389 page records and 96,695 chunks. The page count includes report and box views of shared pages. English report editions fall outside the indexed corpus.

An evidence unit is a text span tied to a stored PDF and its one-based physical page number. Acquisition links catalogue records to source URLs and PDF hashes. Working Paper conversion uses native Poppler text and layout extraction. Report conversion retains structured Docling outputs, element geometry and physical page keys, with a separate native extraction for comparison. Conversion fingerprints identify the derived text. A page citation includes covers in the count and can differ from a printed label. For boxes with verified ownership boundaries, the child record contains the assigned spans. Text with uncertain boundaries remains searchable under the parent. A box and the same passage retrieved through its parent represent one piece of evidence.

3.2. Retrieval and edition scope

The lexical baseline uses weighted BM25 (Robertson and Zaragoza, 2009) over SQLite FTS5 title, abstract and body fields. Metadata and page-bound passage hits are represented separately. The semantic path encodes the query and passage windows with multilingual E5-small (Wang et al., 2024) and scores cosine similarity. When several windows represent a long passage, the system consolidates them at passage level. Hybrid search combines the ranked lists using reciprocal rank fusion (Cormack et al., 2009). Each candidate receives a score

$$s(d) = \sum_{m:\, d \in L_m} \frac{1}{60 + r_m(d)},$$

where L_m is the ranked list returned by retrieval mode m (lexical or semantic) and $r_m(d)$ is the one-based rank. The optional `bge-reranker-v2-m3` cross-encoder reranks windows in the context of candidate pages. It can only rescore pages that reach this stage.

The client accesses the Python retrieval layer through a local Model Context Protocol (MCP) server. It can request context and a rendering of the original page. For edition-specific questions the client can resolve an exact edition or the latest Spanish edition in the local inventory, then search within the report and its boxes. For a historical cutoff the resolver checks the release assertion and evidence that the stored artifact was acquired by the cutoff. Where the evidence is insufficient, it can withhold an as-of selection. This design makes edition scope an explicit input to retrieval and keeps it available for follow-up questions.

3.3. Evaluation design

The first study, conducted on 28 October 2026, used a snapshot containing Working Papers and *Moneda*. Twenty frozen queries were submitted to each of the lexical, semantic and hybrid configurations. Eight prespecified cases also ran with hybrid retrieval and reranking, for 68 calls in total. Each call requested up to five passages. An unblinded reviewer classified first passages as directly useful, partially useful or lacking substantive evidence. The common subset for the four-way comparison consists of six positive natural-language questions. The timing sample for each configuration includes all its calls, including initialization where it occurred.

The second study, audited on 2 November 2026, reviewed sixteen submitted answers from matched keyword and hybrid prompts on eight research questions. Its content gate required full scores for claim support, citation accuracy and uncertainty, a total of at least eight of ten across five dimensions, and zero critical errors. The unblinded audit visually checked 29 selected original PDF pages. The third study, closed on 11 November 2026, examined whether user-supplied first replies in Claude Desktop satisfied a separate answer rubric under frozen instructions. The specified setup used Sonnet High, the local connector and web search disabled. The instructions required resolution of edition scope, physical-page citations and checks on the measure, population, units and period behind numerical claims.

4. Results

4.1. First-passage usefulness and retrieval time

Table 1 reports the first-passage judgments and local retrieval times. Reranking produced substantive first evidence for three of six natural-language questions. One English-language question about climate and financial stability returned a Spanish passage describing financial transmission channels at rank one after reranking. In another cross-language case about the yield curve, a bibliography remained first after reranking and the substantive model description appeared third. For a question about competition and product quality, the relevant page was absent from the candidate pool and could not be recovered by the reranker.

Table 1: First-passage judgments on six natural-language questions. Timing summaries use all calls for each mode.

Mode	Direct	Partial	None	Calls	Median (s)
Keyword	0	0	6	20	0.003
Semantic	0	3	3	20	2.413
Hybrid	0	3	3	20	2.393
Hybrid with reranking	3	2	1	8	31.609

The fixed natural-language wording is consequential for the lexical baseline, whose conjunction of terms can be restrictive. Across all 18 positive queries in the three base modes, the reviewer found six directly useful first passages with keyword search and five each with semantic and hybrid search.

That broader set included exact and filter-oriented queries. The experiment measures sensitivity to the query and ranking configuration. It leaves the effects of deliberate query reformulation, corpus-wide recall and end-to-end client latency unmeasured.

4.2. Evidence use in conversational answers

Four of sixteen answers met the November 2 content gate, and two triggered critical failures. In one case, a cited table and equation used inconsistent coefficient labels. The answer assigned a row to foreign-exchange intervention and drew an inference from its reported significance, although the row-to-variable mapping remained unresolved. In another, the answer reproduced a factor-of-ten error also present in native text extraction. Visual review of the cited PDF page resolved the transcription. These cases show two distinct sources of error after a page has been located: extraction can alter a value, and ambiguity in the underlying publication can prevent interpretation even when the correct page is available.

In the November 11 exercise, the development baseline yielded 11 passes and one partial across 12 answers. The three paired development reruns passed, including the previously partial case. With frozen instructions, eight of eight reserved answers passed with zero critical failures. Edition scope and historical cutoffs passed in applicable cases, including a same-chat follow-up. The evaluation checked selected claims against the local corpus. Client settings and reported page renders were not independently observed, and raw tool-call traces were unavailable. The result establishes performance on this set under the recorded protocol. Different cases and rubrics preclude a before-and-after comparison with the earlier audit, and the absence of a matched held-out control precludes attributing the score to the instructions. End-to-end latency and token usage were unmeasured.

5. Discussion

The experiments suggest that the value of a retrieval method depends on the evidence sought. Lexical search is fast and can be effective when the query contains the source's terminology. Page-context reranking recovered substantive evidence on some natural-language questions at greater computational cost. Its gains are constrained by the candidate set, and similarity scores can return plausible neighbours for unsupported premises. The measured timings reflect local retrieval on the historical snapshot. Client reranking timeouts remain unresolved.

Physical-page provenance makes errors inspectable. Its practical value is greatest when an answer depends on a coefficient, a figure, or a precise claim about what was known by a cutoff. Extraction can degrade equations, tables and reading order, and some JEL metadata remains uncertain. Even a correctly transcribed result needs its population, units, horizon and identification assumptions to travel with it. A larger study could grade evidence at multiple ranks and use fresh cases with independent assessors and raw client traces to distinguish errors in retrieval, interpretation and synthesis.

5.1. Potential research applications

The evaluations cover passage discovery and selected conversational research tasks. Three applications merit further study.

1. **Consistency across long documents.** Compare a figure or definition in an executive summary with the corresponding table, appendix or previous edition. Flag candidate discrepancies with both source pages and check for changes in units, samples or forecast vintages.

2. **Synthesis across large collections.** Assemble evidence for a question whose relevant context is scattered across many reports and papers. For example, reconstruct the public evidence on an inflation shock as of a policy date, preserving the scope of each estimate and exposing missing or conflicting evidence.

3. **Research code generation and replication.** Index approved scripts, variable dictionaries and methodological documentation alongside papers. A client could retrieve the specification, identify documented variables and transformations, and draft code to reproduce a table or adapt an estimation routine. Generated code would require execution and comparison with reference outputs.

5.2. Data governance and deployment

An institutional deployment could incorporate approved internal reports, methodological notes and replication materials if the institution validates the AI client's handling of confidential information. A local retrieval server can still send retrieved passages to the client. Approval would therefore cover the full data path, including who can retrieve material, where prompts and results are processed, and how they are retained or reused.

The portable distribution currently supplies local keyword retrieval through MCP with a separately supplied corpus snapshot. Linux installation success was user-reported. The development updater stages discoveries separately from publication. Its bounded 11 November run recorded no validated additions and made no production change, leaving publication of a real addition, refreshed retrieval and recovery to be validated. The project's personal and internal use scope excludes public redistribution of the PDFs and extracted text.

Planned institutional work begins with a workplace pilot. The proposed shared backend would reuse the retrieval engine and corpus while preserving the local workflow, with a read-only remote MCP interface over Streamable HTTP. A REST interface and container deployment remain options under consideration. Copilot, remote Claude and a research website are prospective clients. Shared use would allow parity, concurrent access and refresh behaviour to be tested against one snapshot, subject to institutional approvals, access controls and document-serving rights.

6. Conclusion

Page-grounded retrieval makes AI-assisted research on central-bank publications easier to audit by keeping passages connected to the artifact and edition that support them. In this corpus, reranking improved first-passage usefulness for some natural-language questions, while answer audits exposed errors that can persist after the relevant page is found. The next empirical question is whether these findings hold across fresh research tasks with independently observable tool use and source review.

Code, data and reproducibility

The public software link in the introduction identifies the portable keyword runtime. Reproducing the October comparison requires the development runtime, the pre-inflation-report database and its vector index. The archived run plan records the frozen queries and fingerprints for the database, index and code. The evaluation record includes passage identities, text hashes and first-passage judgments. The two answer studies retain submitted replies, scoring criteria and selected source-page checks. Exact replay of client behaviour is limited by the missing tool traces and unverified client settings.

The indexed inventory was verified by counting document records by collection and indexing state and counting page and chunk rows. The evaluated snapshots and case-level records are maintained internally. The public software distribution alone is insufficient to reproduce the reported results.

References

Gordon V. Cormack, Charles L. A. Clarke, and Stefan Büttcher. Reciprocal rank fusion outperforms Condorcet and individual rank learning methods. In *Proceedings of the 32nd International ACM SIGIR Conference on Research and Development in Information Retrieval*, pages 758–759, 2009. doi: 10.1145/1571941.1572114.

Leonardo Gambacorta, Byeungchun Kwon, Taejin Park, Pietro Patelli, and Sonya Zhu. CB-LMs: Language models for central banking. BIS Working Paper 1215, Bank for International Settlements, 2024. URL https://www.bis.org/publ/work1215.htm.

Pranab Islam, Anand Kannappan, Douwe Kiela, Rebecca Qian, Nino Scherrer, and Bertie Vidgen. FinanceBench: A new benchmark for financial question answering. arXiv:2311.11944, 2023. URL https://arxiv.org/abs/2311.11944.

Anton Korinek. Generative AI for economic research: Use cases and implications for economists. *Journal of Economic Literature*, 61(4):1281–1317, 2023. doi: 10.1257/jel.20231736.

Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024. doi: 10.1162/tacl_a_00638.

Luca Petracca, Simone De Benedictis, Thomas Gottron, and Zlatina Hofmeister. Supporting users in seeking data on the ECB data portal: A use case for retrieval augmented generation. IFC Bulletin 67, Bank for International Settlements, 2026. URL https://www.bis.org/2026-07/ifcb67_36.pdf.

Stephen Robertson and Hugo Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009. doi: 10.1561/1500000019.

Boon Keong Seah, Eilyn Chong, and Max Chew. Using retrieval-augmented generation (RAG) and LangChain to drive chatbots for interactive economic data analysis and insight. IFC Bulletin 67, Bank for International Settlements, 2026. URL https://www.bis.org/2026-07/ifcb67_37.pdf.

Liang Wang, Nan Yang, Xiaolong Huang, Linjun Yang, Rangan Majumder, and Furu Wei. Multilingual E5 text embeddings: A technical report. arXiv:2402.05672, 2024. URL https://arxiv.org/abs/2402.05672.